MIL-T-60652(MU)
28 June 1968
SUPERSEDING
FA-PD-MI-2690
8 September 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED, M11, REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This specification covers Thruster, Cartridge Actuated, M11, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1167 - Ammunition Data Card

DRAWINGS

Munitions Command
F10521566 - Thruster, Cartridge Actuated, M11, Metal Parts Assembly
C8800969 - Container, Ammunition, Fiber M434 for Thruster, Cartridge Actuated, M11 with Cartridge M141

FSC 1377

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C8800970 - Box, Packing, Ammunition for Thruster, Cartridge Actuated, M11 with Cartridge M141 in Fiber Container, M434

IEL-10521566 - Inspection Equipment List for Thruster, Cartridge Actuated, M11, Metal Parts Assembly

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N. Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Material.- Components for remanufacture may be obtained from thruster models of the XM11 and M11 types.

3.3 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly Drawing F10521566.

3.4 New components.- The remanufactured thruster metal parts assembly shall contain the following new components:

Shear pin	10521574
Curved washer	10521580
Shear screw	10521577
O-ring	MS29513-22, two each
O-ring	MS29513-16
O-ring	MS29513-19, two each
O-ring	MS29513-9

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3.5 Reworked components.

3.5.1 Tube, Inside, C10521578; length of midsection cavity should be 3/8-1/64 long. Pieces with longer lengths shall be modified to meet drawing requirement.

3.5.2 Retainer, C10521575; leading edge of thread lead should have a $30^{\circ} \pm 2^{\circ}$ chamfer. Pieces without the chamfer shall be modified to meet drawing requirement.

3.5.3 Head, F10521569; the large O.D. should have an o-ring groove. Pieces without external o-ring groove shall be modified to meet drawing requirement.

3.6 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied.

3.7 Residual magnetism.- The metal parts assembly shall not deflect the indicator of a pilot's Standby Compass, Type AN5766 more than three degrees in either direction when passed in front of the compass at a distance of 6 inches.

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- All components shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.8.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 and as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lots of metal parts assemblies shall consist of 1200 thrusters, provided the homogeneity requirements of inspection lots have been retained. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed by ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All non-conforming parts shall be either reprocessed or rejected.

4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I and Table II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All non-conforming material shall be rejected.

Table I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1300	100	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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Table II

Total Allowable Defects per Sample Size				
Sample Sizes for Major Defects Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects **Number of Defects in Class**	**110**	**150**	**225**	**300**
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

4.2.2.3 Body, Dwg. D10521568, covering a detail of Dwg. F10521566.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Length of segment groove	Gage
102. Internal chamfer missing or incorrect	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish improper	Visual
106. Burr or sharp corners	Visual
107. Foreign matter	Visual

4.2.2.4 Head, Dwg. F10521569, covering a detail of Dwg. C10521583.

CRITICAL

None defined

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4.2.2.4 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. External o-ring groove	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp edge	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.5 Retainer, Dwg. D10521570, covering a detail of Dwg. C10521575.

CRITICAL	
None defined	
MAJOR	
101. Chamfer missing or incorrect, large end	Visual
102. Thread damaged or not full	Visual
103. Metal defective	Visual
104. Protective finish inadequate or defective	Visual
105. Surface finish on body, improper	Visual
106. Burr or sharp corners	Visual
107. Foreign matter	Visual
MINOR	
None defined	

4.2.2.6 Guide, Dwg. C10521572, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual

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4.2.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Surface finish, improper	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined	

4.2.2.7 Pin, Firing, Dwg. C10521573, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Edge of shear pin hole (No chamfer)	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Surface finish of striker and body, improper	Visual
105. Burr or sharp corners	Visual
106. Foreign matter	Visual
MINOR	
None defined	

4.2.2.8 Ring, Segment, Dwg. C10521576, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Radius or outside surface (No chamfer)	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Surface finish, improper	Visual
105. Burr	Visual
106. Foreign matter	Visual

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4.2.2.8 (Cont'd)

Defect	Method of Inspection
MINOR	
None defined	
4.2.2.9 Tube, Inside, Dwg. C10521578, covering a detail of Dwg. F10521566.	
CRITICAL	
None defined	
MAJOR	
101. Length of midsection cavity	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Surface finish of body and shoulder, improper	Visual
105. Burr or sharp edge	Visual
106. Foreign matter	Visual
MINOR	
None defined	
4.2.2.10 Tube, Outside, Dwg. C10521579, covering a detail of Dwg. F10521566.	
CRITICAL	
None defined	
MAJOR	
101. External o-ring groove	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Surface finish, improper	Visual
105. Burr or sharp edge	Visual
106. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.11 Stop, Dwg. C10521581, covering a detail of Dwg. C10521582.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Width of groove	Gage
102. Total length	Gage
103. Thread damaged or not full	Visual
104. Metal defective	Visual
105. Protective finish inadequate or defective	Visual
106. Surface finish of body and shoulder, improper	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
None defined	

4.2.2.12 Retainer and Pellet Assembly, Dwg. C10521575, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Pellet loose, missing or improperly inserted	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

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4.2.2.13 Stop and Pellet Assembly, Dwg. C10521582, covering a detail of Dwg. F10521566.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Pellet loose, missing or improperly inserted	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.14 Head and Pellet Assembly, Dwg. C10521583, covering a detail of Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Pellet loose, missing or improperly inserted	Visual
102. Thread damaged or not full	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.15 Metal Parts Assembly, Dwg. F10521566, less all components except Guide 10521572 and Firing Pin 10521573.

CRITICAL	
1. Protrusion of firing pin	Gage

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4.2.2.15 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr	Visual
104. Foreign matter	Visual
MINOR	
None defined	

4.2.2.16 Metal parts Assembly, Dwg. F10521566.

CRITICAL	
None defined	
MAJOR	
101. Pellet missing in shear pin hole	Visual
102. Pellet missing in head threads	Visual
103. Improper assembly	Visual
104. Any part missing, not otherwise classified	Visual
105. Material defective	Visual
MINOR	
201. Protective finish defective	Visual
202. Shipping cap missing	Visual
203. Foreign matter	Visual

4.2.2.17 Fiber Container, Dwg. C8800969.

CRITICAL	
None defined	
MAJOR	
None defined	

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4.2.2.17 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual
202. Metal ends loose or distorted	Manual
203. Gap between body and cover, max.	Visual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.18 Packing Box, Dwg. C8800970.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements shall be cause for rejection of the lot.

4.2.3.2 Residual magnetism.- Each assembly shall be subjected to a test for excessive residual magnetism using a Pilot's Standby Compass, Type AN-5766. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the assembly.

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4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL10521566.

4.3 Test methods and procedures.

4.3.1 Protective finish.- Test procedures for protective finish shall be in accordance with the MIL-Standard referenced on the component drawing.

4.3.2 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from, and at the same height as, the compass in a north-south horizontal position with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A - Packing and marking shall be in accordance with Drawings C8800969 and C8800970.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.2 Supersession data.- This specification includes the requirements of Purchase Description FA-PD-MI-2690 dated 8 September 1965.

Custodian:

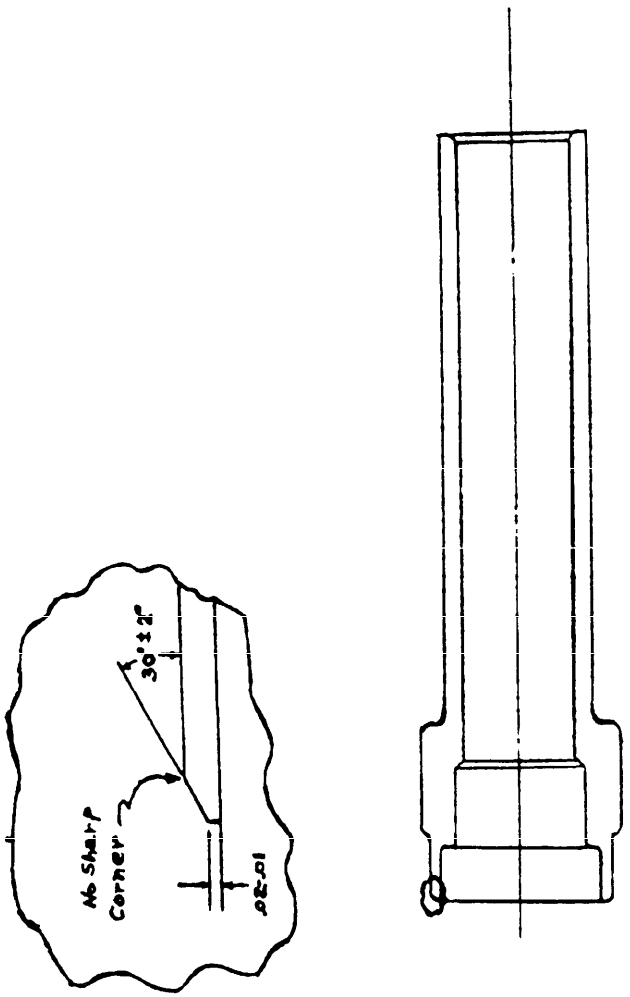
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RETAINER, DWG D10528570

RETAINER MODIFICATION
Date 9-24-65
R.J. Amicone

FIGURE 1.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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